Maximum Likelihood Estimation
&
Maximum A Posteriori Probability Estimation

Outline for today:

1. Maximum Likelihood estimation (MLE)

2. Maximum a posteriori probability (MAP)

3. Example: MLE and MAP for classification

Ex 1: Estimating the probability of a coin flip

We toss a coin n times (independently), we observe the following outcomes:

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Let's make this rigorous!

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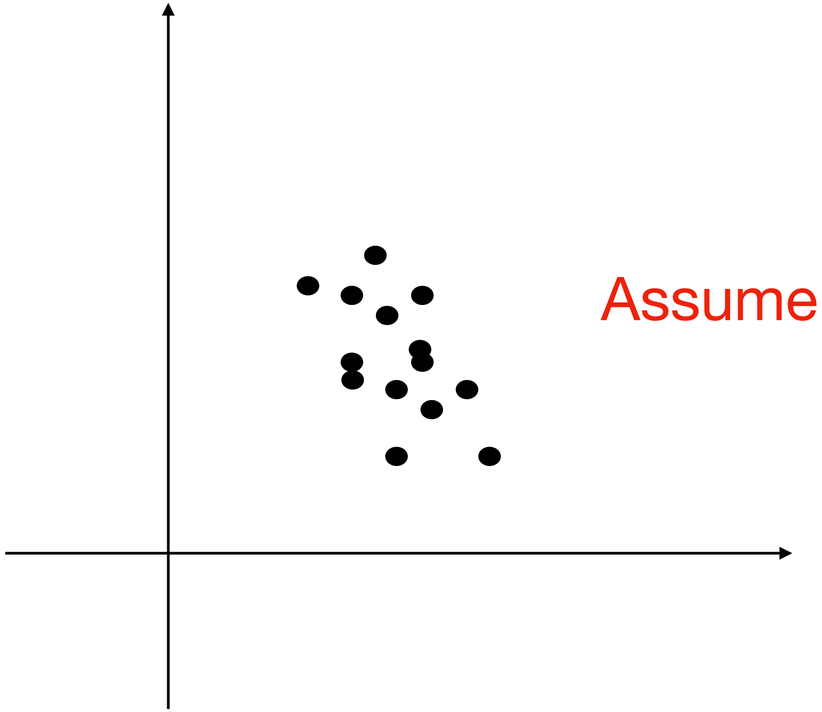

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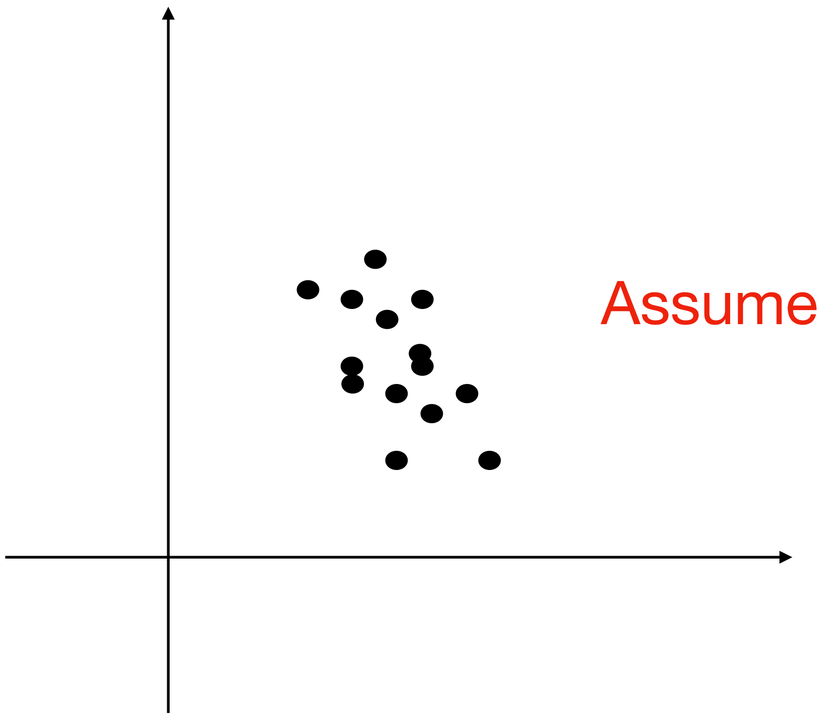

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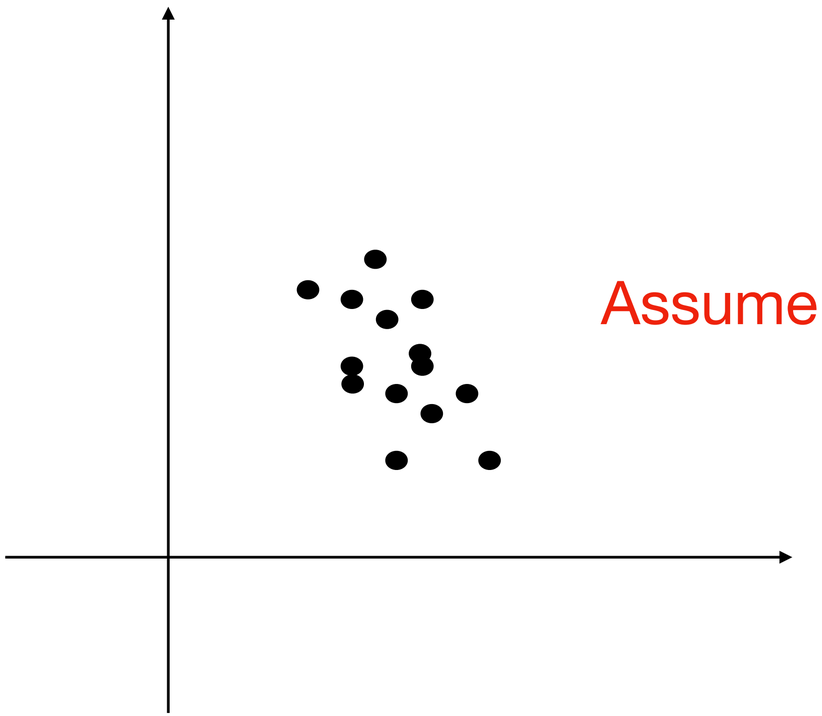

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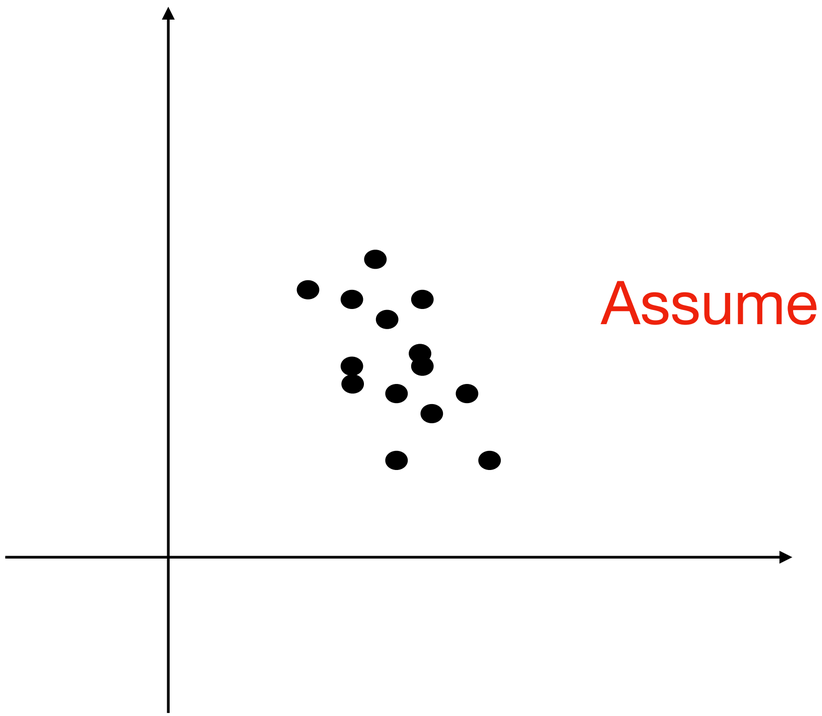

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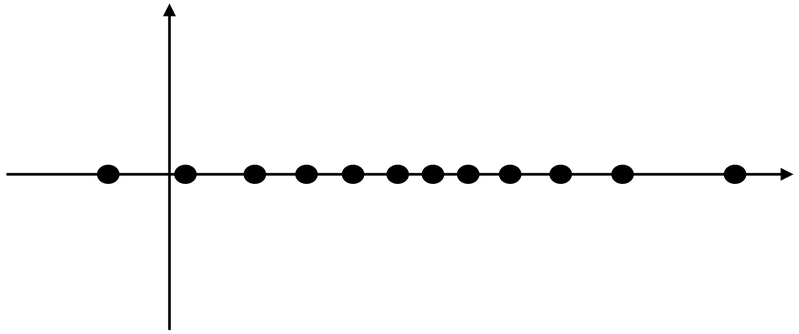

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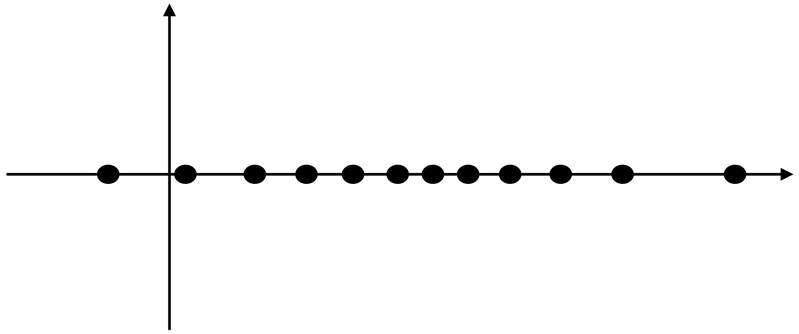

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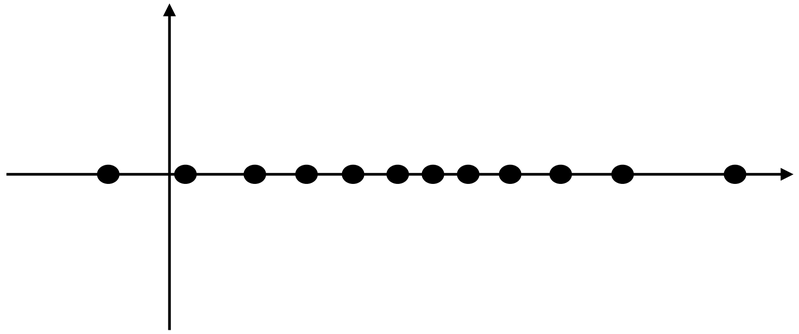

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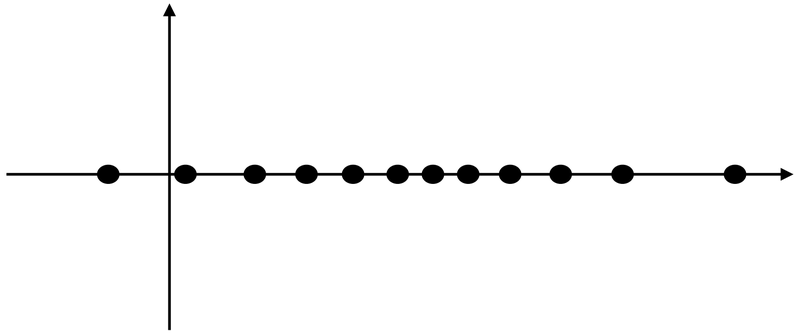

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2. When our model assumption is wrong (e.g., we use Gaussian to model data which is from some more complicated distribution), then MLE loses such guarantee

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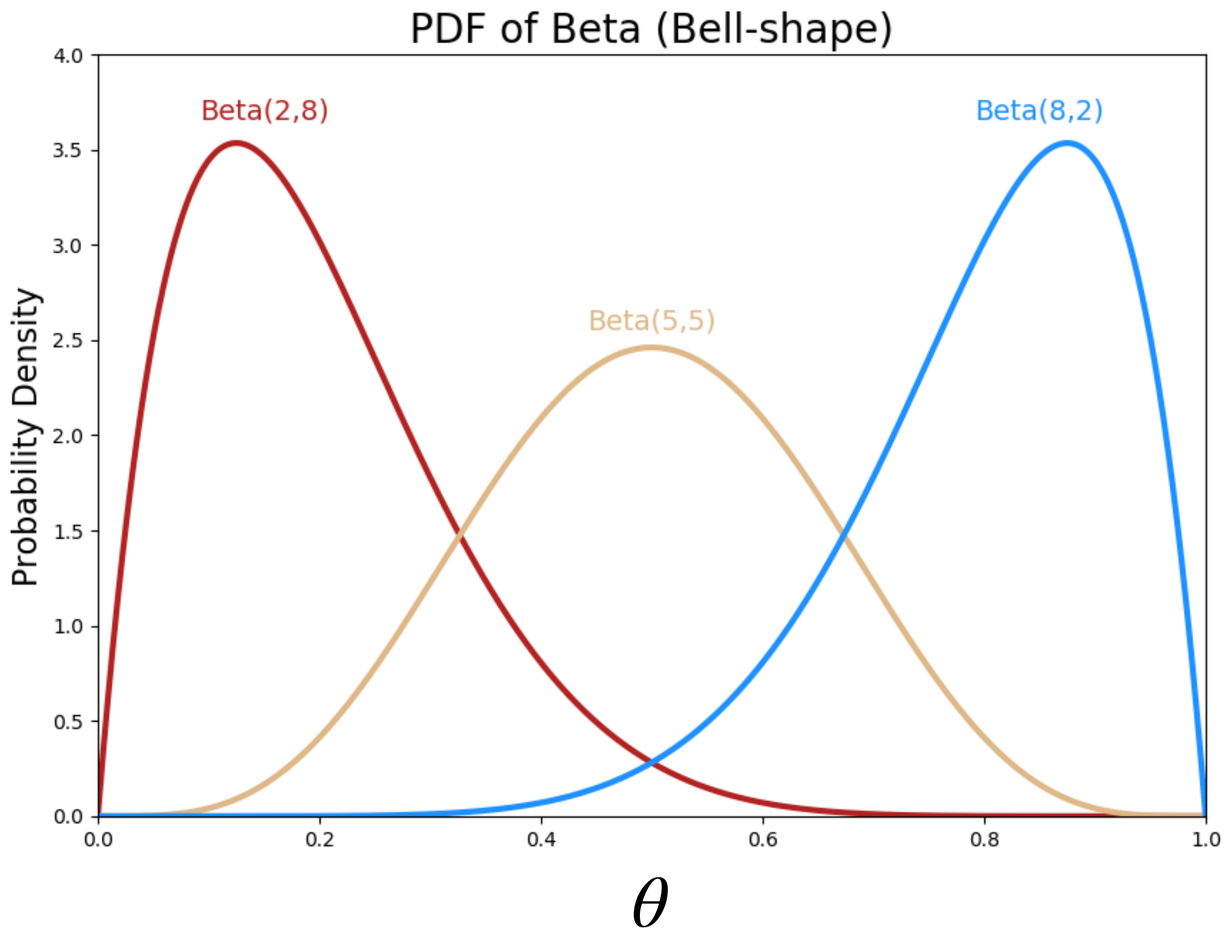

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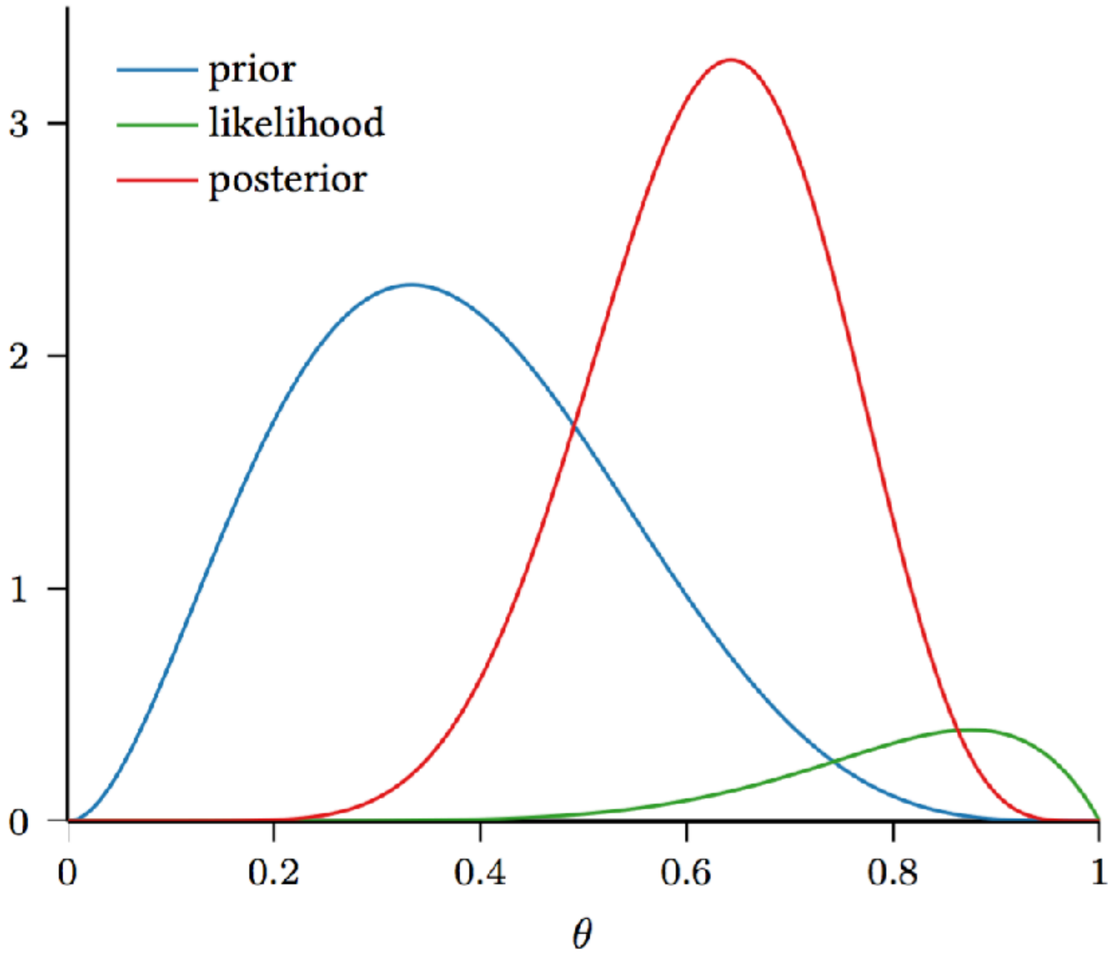

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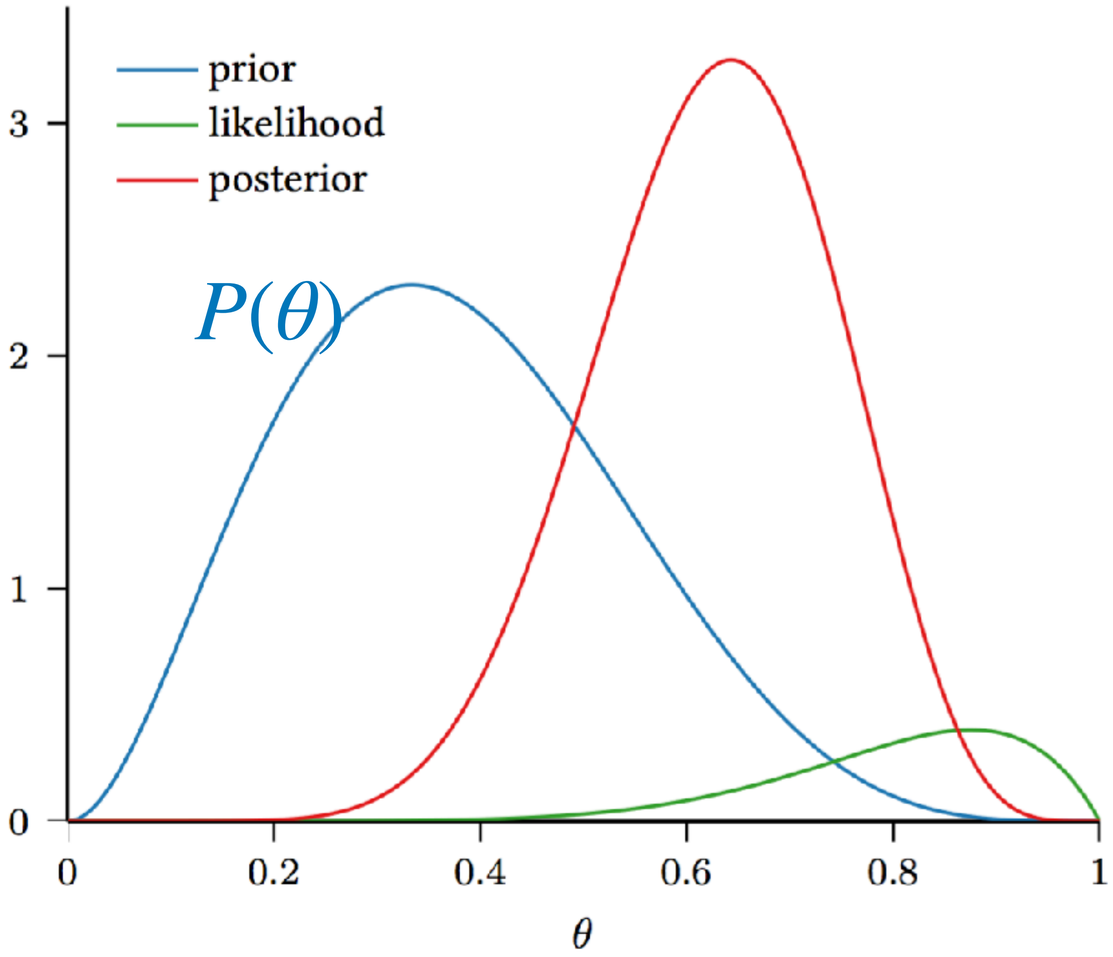

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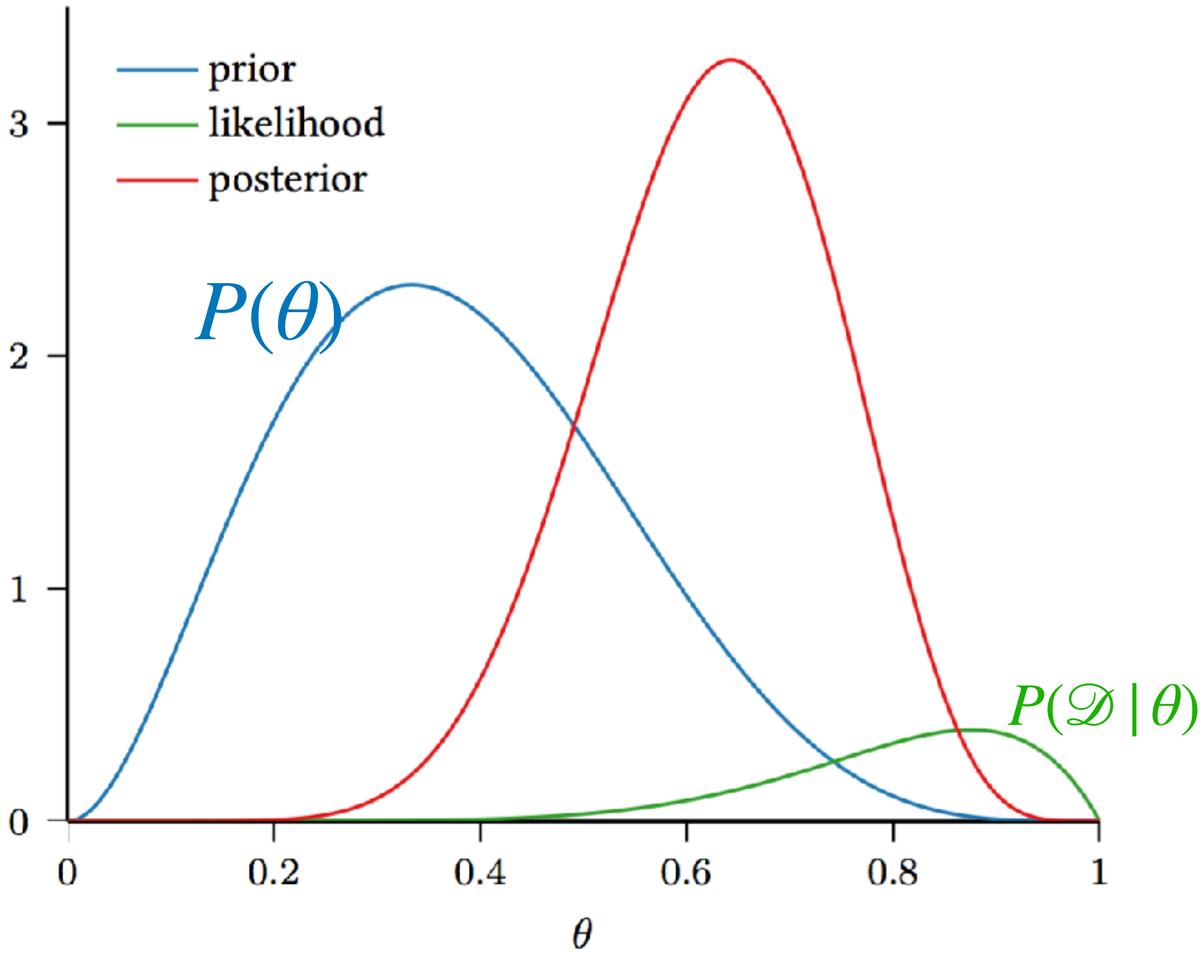

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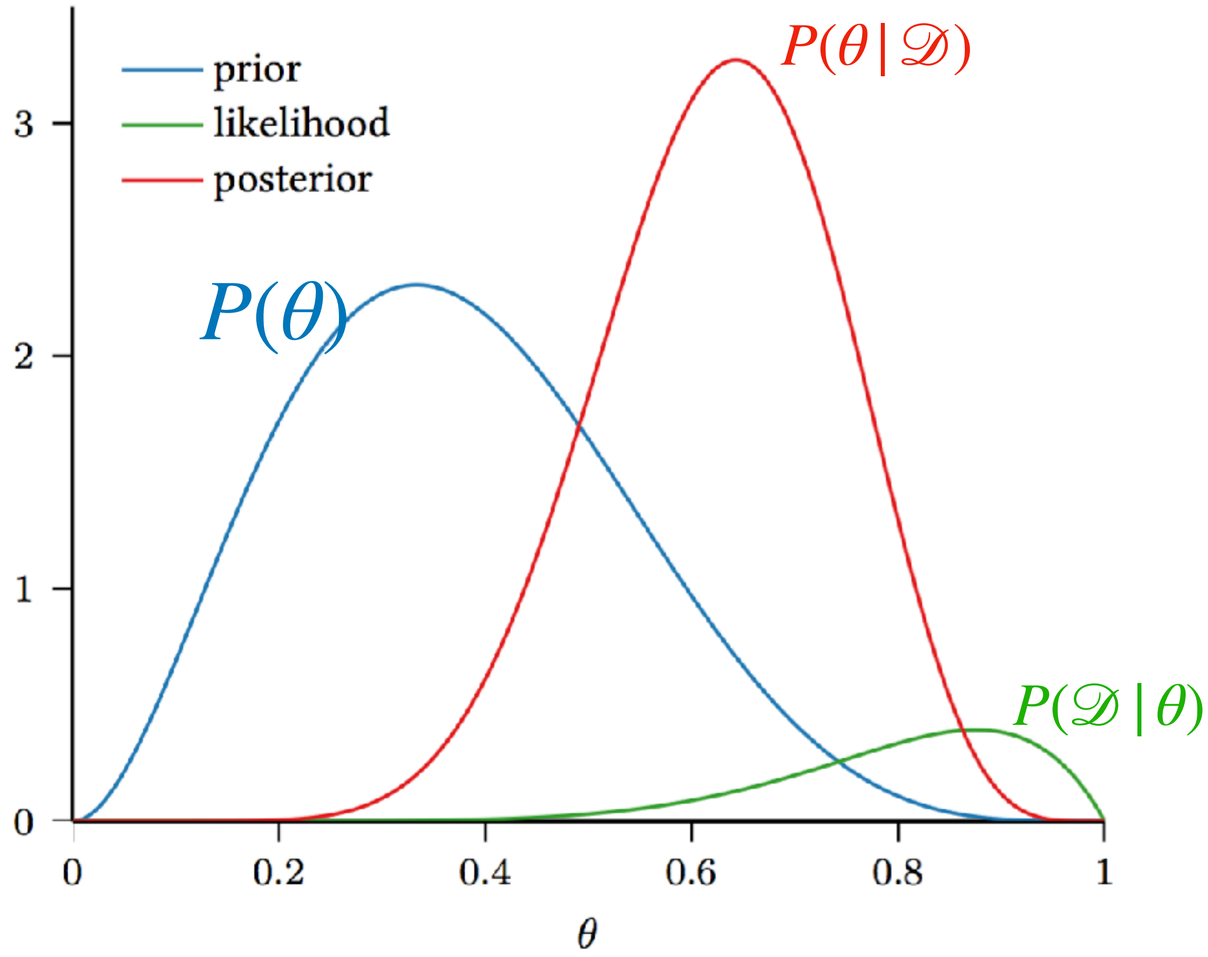

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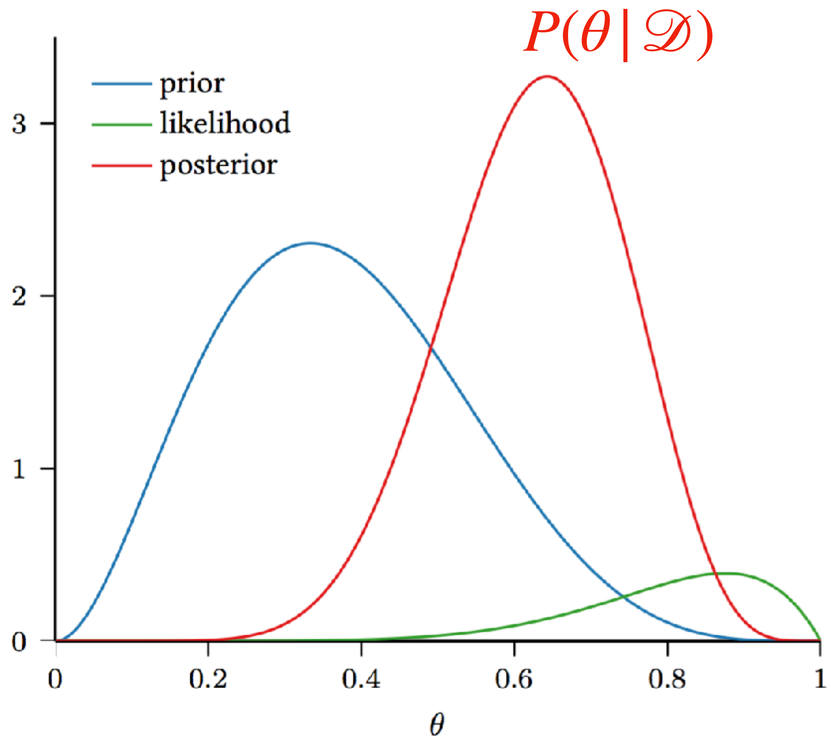

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$(\alpha - 1, \beta - 1)$ can be understood as some fictions flips: we had $\alpha - 1$ hallucinated heads, and $\beta - 1$ hallucinated tails

Some considerations on prior distributions

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3. In general, not so easy to set up a good prior….

Outline for today:

1. Maximum Likelihood estimation (MLE)

2. Maximum a posteriori probability (MAP)

3. Example: MLE and MAP for classification

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Given labeled dataset $\{x_i, y_i\}_{i=1}^n, x_i \in \mathbb{R}^d, y_i \in \{-1,1\}$, we want to estimate $P(y \mid x)$

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Independent of the data

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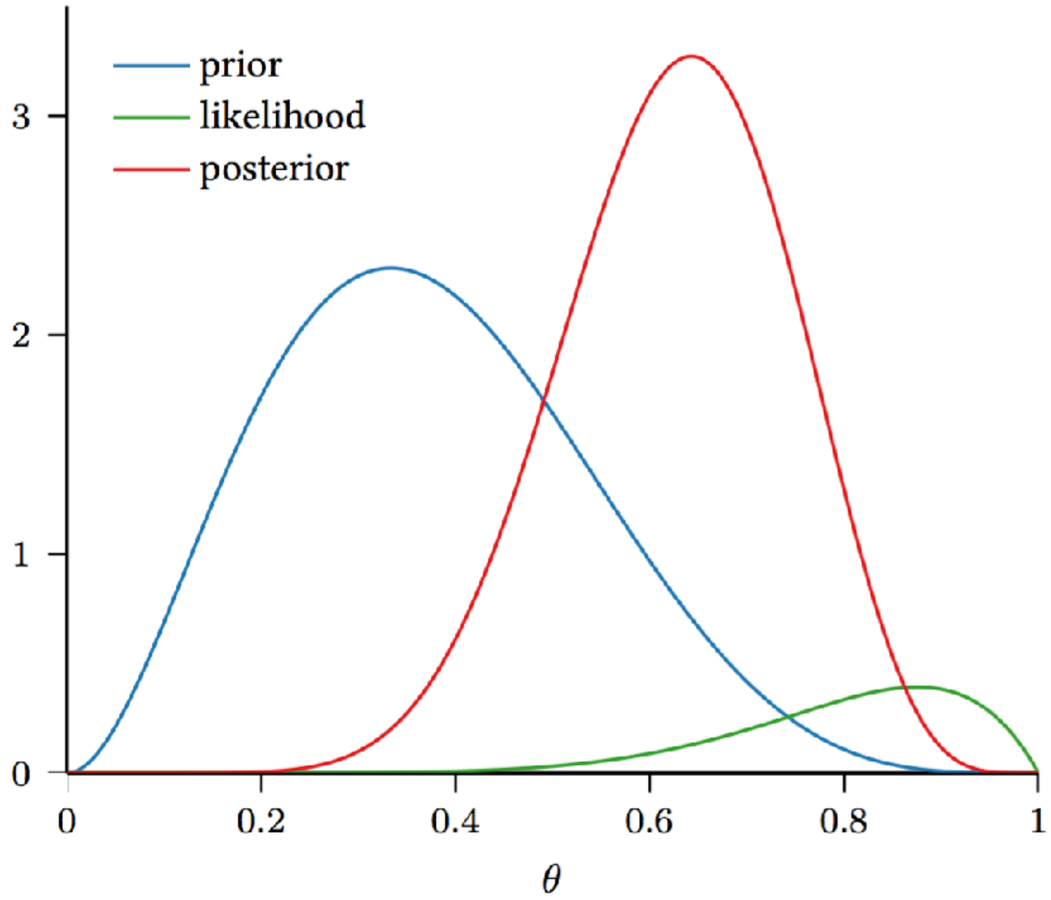

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